

Essential Aspects of Bayesian Data Imputation

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Abstract Data imputation holds significant importance in a variety of fields including risk management. Incomplete or missing data can hinder a thorough analysis of risks, making accurate decision-making challenging. By employing imputation techniques to fill in the gaps, risk managers can obtain a more comprehensive and reliable understanding of the underlying risk factors. This, in turn, enables them to make informed decisions and develop effective strategies for risk mitigation. This note introduces the concept Bayesian data imputation. We collect and provide backgrounds needed for Bayesian data imputation when missing data are missing at random. Numerical examples are provided for demonstration.

Suggested prerequisites. *Having completed undergraduate courses in linear algebra, probability and statistics, as well as a foundational programming course, would be advantageous.*

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1 Introduction

In the current era of rapid technological and scientific advancements, data is generated at an unprecedented rate and has evolved into a fundamental cornerstone of data analysis. Commonly, a data set is fed to a computer and is analyzed through the usage of (statistical) programming languages such as *R*, *Python*. As the analysis crucially depends on the quality of the data set, poor data quality can lead to significant economic costs. It can lead to unfair (potentially wrong) decisions which then could have negative societal impacts [19], [14]. For example, one can use a discretely observed data set to estimate the parameters of a parametric financial model using the maximum likelihood approach. Poor data quality can lead to a significant bias which can affect the parameter estimates of the model.

In the realm of data analysis and machine learning, dealing with missing or incomplete data is a common challenge. Missing data can arise due to a variety of reasons, including measurement errors, equipment malfunctions, or non-response from study participants. For example, [15] reports that R&D information of 42% of all firms is missing between 1980 and 2006. However, working with incomplete data can present difficulties, as it can introduce bias and impede the accuracy of analyses and predictive models. According to [5], the presence of missing data carries several potential implications for empirical tests in risk management, asset pricing and corporate finance. Firstly, excluding firms with missing observations can significantly reduce the data set's size, thereby diminishing the precision of estimates. For instance, it reduces the number of stocks in portfolios constructed by sorting based on characteristics. Moreover, this exclusion can lead to a selection bias if stocks with missing observations differ systematically from those with observed values, especially when the reasons for systematic missingness are not considered. Consequently, asset pricing factor models constructed solely with stocks that possess observed data may overlook crucial systematic pricing information. Secondly, utilizing all available observations but with inaccurately imputed values can introduce a substantial imputation bias, consequently impacting parameter estimates in both structural and reduced-form models.

To tackle the missing data problem, data imputation techniques have emerged as a valuable tool in the field of data analysis. Data imputation is the process of estimating missing values in a data set using available information. Its primary aim is to replace missing values with credible estimates, preserving data integrity and facilitating meaningful analysis. By imputing missing data, researchers and analysts can make more accurate inferences, reduce bias, and improve the performance of statistical models. Various methods exist for data imputation, each with its own strengths and limitations [28], [27]. Traditional imputation techniques include mean imputation, where missing values are replaced with the mean of observed values for that particular variable, or regression imputation, where missing values are estimated using a regression model based on other variables. While these methods are relatively simple, they may introduce biases or underestimate the uncertainty associated with imputed values.

In recent years, more sophisticated imputation techniques leveraging machine learning algorithms have been developed [18]. Examples of these methods include k -nearest neighbors imputation [3], which imputes missing values based on the values of the nearest neighbors in the feature space, and multiple imputation [30], which generates multiple plausible imputed data sets to capture the uncertainty associated with imputation.

Data imputation is a critical step in the data preprocessing pipeline and plays a crucial role in various fields, including healthcare, social sciences, finance, and customer analytics. By salvaging valuable information that would have otherwise been lost due to missing data, imputation allows researchers and analysts to conduct more robust and reliable analyses. Research in this direction has attracted a great deal of attention from researchers in a wide range of disciplines including mathematics, statistics, and computer science. Many textbooks have been written about this topic: [27], [29], [19]. Various software packages which can help user to efficiently deal with missing values have been introduced. For instance, the R package *mice*, as described in [30], offers users the capability to impute incomplete multivariate data through chained equations. It exhibits high flexibility, enabling the handling of variables with different types (e.g., continuous or binary), and can accommodate complexities such as bounds and survey skip patterns. One can refer to [1] to see the ideas behind the multivariate imputation by chained equations. Recently, ones are interested in using machine learning techniques to deal with missing data [4]. MIDAS (Multiple Imputation with Denoising Autoencoders), for instance, uses denoising autoencoders, a type of unsupervised neural network, to learn fine-grained latent representations of the observed data [18]. MIDAS has been shown to provide accuracy and efficiency advantages over traditional multiple imputation strategies.

The development of Bayesian data analysis has been a significant milestone in the field of statistics and decision-making [23], [12], [17]. While its foundations can be traced back to the 18th-century work of the Reverend Thomas Bayes [2], it was not until the late 20th century that the field witnessed remarkable advancements. Bayesian data analysis is built upon the principles of Bayes' theorem, offering a framework for updating our beliefs about a hypothesis or model as new data becomes available. This approach allows for the seamless integration of prior knowledge or assumptions with observed data, resulting in more nuanced and flexible analyses. Over time, computational advancements and the accessibility of powerful computing resources have further propelled the development of Bayesian data analysis. Today, Bayesian methods find wide application across diverse disciplines, including medicine, economics, social sciences, and machine learning. They provide a robust and principled framework for inference, prediction, and decision-making in the face of uncertainty [10].

In this note, we collect and provide a Bayesian perspective on data imputation. Our note is inspired by Chapter 7 of [12]. Our goal is to provide a self-contained introduction to Bayesian data imputation which we think would greatly benefit undergraduates. The rest of this chapter is organized as follows: In Section 2 we recall some basic definitions of probability and statistics needed for later sections. Section 3 provides the preliminary work needed for Section 4. In particular, Section 3.1 re-

calls the key concepts of Bayesian inference. Section 4 collects the main results of the chapter where both theoretical foundations and several numerical examples are provided. Section 5 concludes the chapter.

2 Multivariate Normal Distribution

In probability theory and statistics, the multivariate normal distribution, also known as the multivariate Gaussian distribution, extends the one-dimensional (univariate) normal distribution to higher dimensions. This distribution is commonly employed for modeling high-dimensional data sets.

We will begin by revisiting some fundamental properties of the multivariate normal distribution, including its probability density function, maximum likelihood estimations, and its conditional distribution. These concepts will be extensively utilized in subsequent sections.

2.1 Probability Density Function

Let $\mathbf{X} = (X_1, X_2, \dots, X_p)^T \in \mathbb{R}^p$ be a p -dimensional random vector. Also, let $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_p)^T \in \mathbb{R}^p$ and $\boldsymbol{\Sigma} = (\Sigma_{ij})_{p \times p} \in \mathbb{R}^{p \times p}$ be a symmetric positive definite matrix. It is said that $\mathbf{X}$ has a multivariate-normal distribution with the parameters $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$ if the probability density function of $\mathbf{X}$ is given by

$$p(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\theta})\right), \quad \mathbf{x} \in \mathbb{R}^p.$$

Note that in the above equation, $\boldsymbol{\Sigma}^{-1}$ denotes the matrix inverse of $\boldsymbol{\Sigma}$. That is $\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1} = \mathbf{I}_{p \times p} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}$. Also, $|\boldsymbol{\Sigma}|$ denotes the determinant of the matrix $\boldsymbol{\Sigma}$. In fact, we have $\boldsymbol{\theta} = \mathbb{E}[\mathbf{X}] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_p])^T$ and $\boldsymbol{\Sigma} = \mathbb{E}[(\mathbf{X} - \boldsymbol{\theta})(\mathbf{X} - \boldsymbol{\theta})^T]$. Concretely, $\Sigma_{ij} = \mathbb{E}[(X_i - \theta_i)(X_j - \theta_j)] = \text{Cov}[X_i, X_j]$, $1 \leq i, j \leq p$. See Theorem 1 below.

The quantity $\sqrt{(\mathbf{x} - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\theta})}$ is known as the Mahalanobis distance [6], which represents the distance of the test point $\mathbf{x}$ from the mean $\boldsymbol{\theta}$. In the case when $p = 1$, the distribution reduces to a univariate normal distribution and the Mahalanobis distance reduces to the absolute value of the standard score. Moreover, the shape of the density function is determined by the quadratic form $(\mathbf{x} - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\theta})$ whose level surfaces are ellipsoids oriented along the eigenvectors of the matrix $\boldsymbol{\Sigma}$. Since $\boldsymbol{\Sigma}$ is positive definite, its inverse $\boldsymbol{\Sigma}^{-1}$ is also positive definite. As a result, the paraboloid $(\mathbf{x} - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\theta})$ has a minimum at $\mathbf{x} = \boldsymbol{\theta}$. Therefore, $p(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\Sigma})$ has a “bump” in $(p + 1)$ -dimensional space centered at $\mathbf{x} = \boldsymbol{\theta}$. In Figure 1, we plot the contours of the probability density function of $p(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\Sigma})$ with $\boldsymbol{\theta} = (0, 0)^T$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (Case I), $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$ (Case II). As we can see from Figure 1, the

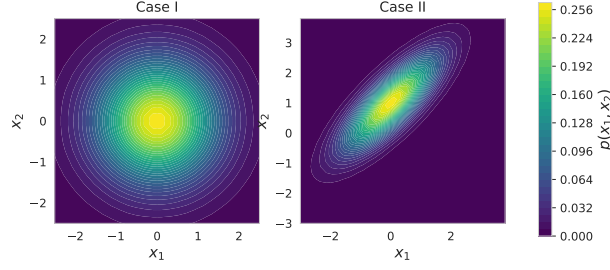


Fig. 1 Bivariate multivariate normal distributions

shape of the distribution is determined entirely by the covariance matrix Σ . One can refer to the book [31] for more properties of the multivariate normal distributions. In this chapter, we will adopt the following notation

$$p(\mathbf{x}|\boldsymbol{\theta}, \Sigma) \propto \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\theta})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\theta})\right).$$

When $\mathbf{X} = (X_1, X_2, \dots, X_p)^T$ has a multivariate-normal distribution with parameters $\boldsymbol{\theta}$ and Σ , we will denote it as

$$\mathbf{X} \sim \text{multivariate-normal}(\boldsymbol{\theta}, \Sigma).$$

In case the covariance matrix Σ is a symmetric positive definite diagonal matrix, $\Sigma = \text{diag}(\Sigma_1^2, \Sigma_2^2, \dots, \Sigma_p^2)$, it can be seen immediately that

$$\begin{aligned} p(\mathbf{x}|\boldsymbol{\theta}, \Sigma) &= \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\theta})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\theta})\right) \\ &= \prod_{i=1}^p \frac{1}{\Sigma_i \sqrt{2\pi}} \exp\left(-\frac{1}{2\Sigma_i^2}(x_i - \theta_i)^2\right). \end{aligned}$$

As a result $X_1, X_2, \dots, X_p$ are independent random variables. It is immediate to see that a univariate normal random variable is obtained when $p = 1$. The following theorem is well known. We however could not locate a direct proof in standard textbooks such as [31]. Hence, we collect it here for the sake of completeness.

Theorem 1 If $\mathbf{X} \sim \text{multivariate-normal}(\boldsymbol{\theta}, \Sigma)$ then

$$\mathbb{E}(\mathbf{X}) = \boldsymbol{\theta} \quad \text{and} \quad \text{Cov}(\mathbf{X}) = \Sigma.$$

Proof. Note that since Σ is positive definite, from [9] we have $\Sigma = \mathbf{U}\mathbf{V}\mathbf{U}^T$ where $\mathbf{U}$ is a full rank orthogonal matrix containing of the eigenvectors of Σ as its columns and $\mathbf{V} = \text{diag}(\lambda_1, \dots, \lambda_p)$ is a diagonal matrix containing Σ 's eigenvalues $\lambda_i > 0, i = 1, 2, \dots, p$. Let the matrix $\mathbf{V}^{1/2} = \text{diag}(\lambda_1^{1/2}, \dots, \lambda_p^{1/2})$ and $\mathbf{B} = \mathbf{U}\mathbf{V}^{1/2}$. It is immediate to check that $\Sigma = \mathbf{B}\mathbf{B}^T$. Hence $\Sigma^{-1} = \mathbf{B}^{-T}\mathbf{B}^{-1}$. As a result,

$$\begin{aligned}
\mathbb{E}(\mathbf{X}) &= \int_{\mathbb{R}^p} \mathbf{x} p(\mathbf{x} | \boldsymbol{\theta}, \boldsymbol{\Sigma}) d\mathbf{x} \\
&= \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \int_{\mathbb{R}^p} \mathbf{x} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\theta})\right) d\mathbf{x} \\
&= \frac{1}{(2\pi)^{p/2} |\mathbf{B}\mathbf{B}^T|^{1/2}} \int_{\mathbb{R}^p} \mathbf{x} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\theta})^T \mathbf{B}^{-T} \mathbf{B}^{-1}(\mathbf{x} - \boldsymbol{\theta})\right) d\mathbf{x} \\
&= \frac{1}{(2\pi)^{p/2} |\mathbf{B}|} \int_{\mathbb{R}^p} \mathbf{x} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\theta})^T \mathbf{B}^{-T} \mathbf{B}^{-1}(\mathbf{x} - \boldsymbol{\theta})\right) d\mathbf{x}.
\end{aligned}$$

Now let $\mathbf{z} = \mathbf{B}^{-1}(\mathbf{x} - \boldsymbol{\theta})$, this implies that $\mathbf{x} = \mathbf{B}\mathbf{z} + \boldsymbol{\theta}$. Therefore

$$\begin{aligned}
\mathbb{E}(\mathbf{X}) &= \frac{1}{(2\pi)^{p/2} |\mathbf{B}|} \int_{\mathbb{R}^p} \mathbf{x} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\theta})^T \mathbf{B}^{-T} \mathbf{B}^{-1}(\mathbf{x} - \boldsymbol{\theta})\right) d\mathbf{x} \\
&= \frac{1}{(2\pi)^{p/2} |\mathbf{B}|} \int_{\mathbb{R}^p} (\mathbf{B}\mathbf{z} + \boldsymbol{\theta}) \exp\left(-\frac{1}{2}\mathbf{z}^T \mathbf{z}\right) |\mathbf{B}| d\mathbf{z} \\
&= \frac{1}{(2\pi)^{p/2} |\mathbf{B}|} \int_{\mathbb{R}^p} \mathbf{B}\mathbf{z} \exp\left(-\frac{1}{2}\mathbf{z}^T \mathbf{z}\right) |\mathbf{B}| d\mathbf{z} + \frac{1}{(2\pi)^{p/2} |\mathbf{B}|} \int_{\mathbb{R}^p} \boldsymbol{\theta} \exp\left(-\frac{1}{2}\mathbf{z}^T \mathbf{z}\right) |\mathbf{B}| d\mathbf{z} \\
&= \frac{1}{(2\pi)^{p/2} |\mathbf{B}|} \int_{\mathbb{R}^p} \boldsymbol{\theta} \exp\left(-\frac{1}{2}\mathbf{z}^T \mathbf{z}\right) |\mathbf{B}| d\mathbf{z} \\
&= \boldsymbol{\theta} \prod_{i=1}^p \frac{1}{(2\pi)^{1/2}} \int_{\mathbb{R}} \exp\left(-\frac{1}{2}z_i^2\right) dz_i \\
&= \boldsymbol{\theta}.
\end{aligned}$$

From the substitution $\mathbf{Z} = \mathbf{B}^{-1}(\mathbf{X} - \boldsymbol{\theta})$ and the calculation above we can see that the probability density function of $\mathbf{Z} = (Z_1, \dots, Z_p)$ is given by

$$\begin{aligned}
p(\mathbf{z}) &= \frac{1}{(2\pi)^{p/2}} \exp\left(-\frac{1}{2}\mathbf{z}^T \mathbf{z}\right), \quad \mathbf{z} \in \mathbb{R}^p \\
&= \prod_{i=1}^p \frac{1}{(2\pi)^{1/2}} \exp\left(-\frac{1}{2}z_i^2\right) \\
&= p(z_1) \dots p(z_p),
\end{aligned}$$

where $p(z_i)$ the the probability density function of $Z_i, i = 1, 2, \dots, p$. As a result, the random variables $Z_1, \dots, Z_p$ are independent. Moreover, $Z_i \sim N(0, 1)$. Therefore,

$$\mathbf{Z} \sim \text{multivariate-normal}(\mathbf{0}, \mathbf{I})$$

Hence

$$\text{Cov}(\mathbf{X}) = \text{Cov}(\mathbf{B}\mathbf{Z} + \boldsymbol{\theta}) = \mathbf{B} \text{Cov}(\mathbf{Z}) \mathbf{B}^T = \mathbf{B}\mathbf{B}^T = \boldsymbol{\Sigma}.$$

This completes the proof of the theorem. □

Exercise 1. Assume that $-1 < \rho < 1$ and $\mathbf{X} = (X_1, X_2)^T$ is normally distributed with mean $\boldsymbol{\theta} = (\theta_1, \theta_2)^T$ and covariance $\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$.

1. Show that $|\boldsymbol{\Sigma}| = \sigma_1^2\sigma_2^2(1 - \rho^2)$.
2. Find $\boldsymbol{\Sigma}^{-1}$.
3. Show that the joint probability density function of $\mathbf{X} = (X_1, X_2)^T$ can be written as

$$p(x_1, x_2 | \boldsymbol{\theta}, \boldsymbol{\Sigma}) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \times \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x_1 - \theta_1}{\sigma_1}\right)^2 - 2\rho \left(\frac{x_1 - \theta_1}{\sigma_1}\right) \left(\frac{x_2 - \theta_2}{\sigma_2}\right) + \left(\frac{x_2 - \theta_2}{\sigma_2}\right)^2 \right]\right).$$

2.2 Maximum Likelihood Estimation

Assume that an iid sample $\{\mathbf{x}_i : i = 1, 2, \dots, n\}$ is collected and $\mathbf{x}_i \sim \text{multivariate-normal}(\boldsymbol{\theta}, \boldsymbol{\Sigma})$, $i = 1, 2, \dots, n$ with $n > p$. We ask for estimators of the mean vector $\boldsymbol{\theta}$ and the covariance matrix $\boldsymbol{\Sigma}$ of the distribution. It turns out that we can use the maximum likelihood estimation (MLE) to estimate $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$ from the collected sample as follows.

Theorem 2 *If we have an iid sample $\{\mathbf{x}_i \sim \text{multivariate-normal}(\boldsymbol{\theta}, \boldsymbol{\Sigma}) : i = 1, 2, \dots, n\}$, then the MLE for the parameters $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$ are, respectively, given by*

$$\hat{\boldsymbol{\theta}}_{mle} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \equiv \bar{\mathbf{x}},$$

$$\hat{\boldsymbol{\Sigma}}_{mle} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T.$$

Proof. First, the likelihood function can be found as

$$\begin{aligned} p(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n | \boldsymbol{\theta}, \boldsymbol{\Sigma}) &= \prod_{i=1}^n \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\theta})\right) \\ &= \frac{1}{(2\pi)^{np/2} |\boldsymbol{\Sigma}|^{n/2}} \exp\left(-\frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\theta})\right). \end{aligned}$$

Let $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$, then the log-likelihood function is given by

$$\begin{aligned} \log(\boldsymbol{\theta}, \boldsymbol{\Sigma}) &= \log(p(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n | \boldsymbol{\theta}, \boldsymbol{\Sigma})) \\ &= \log((2\pi)^{-np/2}) + \frac{n}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T \boldsymbol{\Lambda}(\mathbf{x}_i - \boldsymbol{\theta}). \end{aligned}$$

Therefore,

$$\begin{aligned}
 \frac{\partial}{\partial \boldsymbol{\theta}} \log(\boldsymbol{\theta}, \boldsymbol{\Sigma}) &= \frac{\partial}{\partial \boldsymbol{\theta}} \left(-\frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T \boldsymbol{\Lambda} (\mathbf{x}_i - \boldsymbol{\theta}) \right) \\
 &= -\frac{1}{2} \sum_{i=1}^n -2\boldsymbol{\Lambda} (\mathbf{x}_i - \boldsymbol{\theta}) \\
 &= \boldsymbol{\Lambda} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta}).
 \end{aligned}$$

Hence

$$\begin{aligned}
 \frac{\partial}{\partial \boldsymbol{\theta}} \log(\boldsymbol{\theta}, \boldsymbol{\Sigma}) &= 0 \\
 \Leftrightarrow \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta}) &= 0 \\
 \Leftrightarrow \sum_{i=1}^n \mathbf{x}_i - \sum_{i=1}^n \boldsymbol{\theta} &= 0 \\
 \Leftrightarrow \sum_{i=1}^n \mathbf{x}_i - n\boldsymbol{\theta} &= 0 \\
 \Leftrightarrow \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i &= \boldsymbol{\theta}.
 \end{aligned}$$

Next, the trace of a matrix $\mathbf{A} = (a_{ij})$ is denote by $tr(\mathbf{A}) = \sum_i a_{ii}$. We recall the following property of trace: For matrices of appropriate sizes $\mathbf{A}, \mathbf{B}, \mathbf{C}$ it is well-known (see [24]) that

$$tr(\mathbf{ABC}) = tr(\mathbf{CAB}) = tr(\mathbf{BCA}).$$

Also, for any vector $\mathbf{x}$ and matrix $\mathbf{A}$ of appropriate sizes, from the above result we have

$$\mathbb{R} \ni \mathbf{x}^T \mathbf{A} \mathbf{x} = tr(\mathbf{x}^T \mathbf{A} \mathbf{x}) = tr(\mathbf{x} \mathbf{x}^T \mathbf{A}) = tr(\mathbf{A} \mathbf{x} \mathbf{x}^T).$$

As a result, we can rewrite the log-likelihood function as

$$\begin{aligned}
\log(\boldsymbol{\theta}, \boldsymbol{\Sigma}) &= \log((2\pi)^{-np/2}) + \frac{n}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T \boldsymbol{\Lambda} (\mathbf{x}_i - \boldsymbol{\theta}) \\
&= \log((2\pi)^{-np/2}) + \frac{n}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \sum_{i=1}^n \text{tr}[(\mathbf{x}_i - \boldsymbol{\theta})^T \boldsymbol{\Lambda} (\mathbf{x}_i - \boldsymbol{\theta})] \\
&= \log((2\pi)^{-np/2}) + \frac{n}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \sum_{i=1}^n \text{tr}[(\mathbf{x}_i - \boldsymbol{\theta})^T (\mathbf{x}_i - \boldsymbol{\theta}) \boldsymbol{\Lambda}] \\
&= \log((2\pi)^{-np/2}) + \frac{n}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \text{tr}[\sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T (\mathbf{x}_i - \boldsymbol{\theta}) \boldsymbol{\Lambda}] \\
&= \log((2\pi)^{-np/2}) + \frac{n}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \text{tr}[S_{\boldsymbol{\theta}} \boldsymbol{\Lambda}],
\end{aligned}$$

where $S_{\boldsymbol{\theta}} = \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T (\mathbf{x}_i - \boldsymbol{\theta})$.

Next, for any two matrices $\mathbf{A}$ and $\mathbf{B}$ of the appropriate sizes, we have

$$\frac{\partial}{\partial a_{ij}} \text{tr}(\mathbf{AB}) = \frac{\partial}{\partial a_{ij}} \sum_k \sum_l a_{kl} b_{lk} = b_{ji},$$

which shows that

$$\frac{\partial}{\partial \mathbf{A}} \text{tr}(\mathbf{AB}) = \mathbf{B}^T.$$

Moreover, for a matrix $\mathbf{A}$ such that $|\mathbf{A}| \neq 0$, let $\mathbf{M}_{ij}$ be the corresponding minor and $\tilde{\mathbf{A}}$ be the matrix of cofactors. We have

$$\begin{aligned}
\frac{\partial}{\partial a_{ij}} \log(|\mathbf{A}|) &= \frac{1}{|\mathbf{A}|} \frac{\partial}{\partial a_{ij}} |\mathbf{A}| \\
&= \frac{1}{|\mathbf{A}|} \frac{\partial}{\partial a_{ij}} \sum_j (-1)^{i+j} a_{ij} M_{ij} \\
&= \frac{1}{|\mathbf{A}|} \sum_j (-1)^{i+j} M_{ij} \\
&= \frac{1}{|\mathbf{A}|} \tilde{\mathbf{A}}_{ij}^T \\
&= \frac{1}{|\mathbf{A}^T|} \tilde{\mathbf{A}}_{ij}^T.
\end{aligned}$$

The above calculation shows that

$$\frac{\partial}{\partial \mathbf{A}} \log(|\mathbf{A}|) = \mathbf{A}^{-T}.$$

Therefore,

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{\Lambda}} \log(\boldsymbol{\theta}, \boldsymbol{\Sigma}) &= \frac{n}{2} \mathbf{\Lambda}^{-T} - \frac{1}{2} S_{\boldsymbol{\theta}}^T = 0 \\
&\Leftrightarrow n \mathbf{\Lambda}^{-1} - S_{\boldsymbol{\theta}}^T = 0 \\
&\Leftrightarrow \mathbf{\Lambda}^{-1} - \frac{1}{n} S_{\boldsymbol{\theta}}^T = 0 \\
&\Leftrightarrow \boldsymbol{\Sigma} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})(\mathbf{x}_i - \boldsymbol{\theta})^T = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T.
\end{aligned}$$

This completes the proof of the theorem. □

We note that the MLE estimates for $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$ will be useful later when one needs to initialize the Gibbs sampling algorithm. Please see Section 4 for more details.

Exercise 2. Prove that

$$\frac{\partial}{\partial \mathbf{\Lambda}} \left(\frac{n}{2} \log |\mathbf{\Lambda}| - \frac{1}{2} \text{tr}[S_{\boldsymbol{\theta}} \mathbf{\Lambda}] \right) = \frac{n}{2} \mathbf{\Lambda}^{-T} - \frac{1}{2} S_{\boldsymbol{\theta}}^T.$$

2.3 Conditional Distribution

In this section, we assume that $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2) \in \mathbb{R}^{d_1 \times d_2}$ is jointly normally distributed multivariate-normal($\boldsymbol{\theta}, \boldsymbol{\Sigma}$) with parameters

$$\boldsymbol{\theta} = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}.$$

We are interested in the conditional distribution $p(\mathbf{x}_1 | \mathbf{x}_2)$. This will provide the key equation for a Bayesian data imputation developed in Section 4. We have the following theorem regarding the closed form distribution of $p(\mathbf{x}_1 | \mathbf{x}_2)$.

Theorem 3 *The conditional distribution $p(\mathbf{x}_1 | \mathbf{x}_2)$ is given by*

$$p(\mathbf{x}_1 | \mathbf{x}_2) \sim \text{multivariate-normal}(\boldsymbol{\theta}_{1|2}, \boldsymbol{\Sigma}_{1|2})$$

where

$$\boldsymbol{\theta}_{1|2} = \boldsymbol{\theta}_1 + \Sigma_{12} \Sigma_{22}^{-1} (\mathbf{x}_2 - \boldsymbol{\theta}_2), \quad \boldsymbol{\Sigma}_{1|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}.$$

Proof. Recall that the joint probability density function of $(\mathbf{X}_1, \mathbf{X}_2)$ can be written as

$$p(\mathbf{x}_1, \mathbf{x}_2 | \boldsymbol{\theta}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{(d_1+d_2)/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} \begin{pmatrix} \mathbf{x}_1 - \boldsymbol{\theta}_1 \\ \mathbf{x}_2 - \boldsymbol{\theta}_2 \end{pmatrix}^T \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{x}_1 - \boldsymbol{\theta}_1 \\ \mathbf{x}_2 - \boldsymbol{\theta}_2 \end{pmatrix} \right).$$

Note that one can directly obtain the following identity,

$$\begin{pmatrix} \mathbf{I} - \Sigma_{12}\Sigma_{22}^{-1} & \\ 0 & \mathbf{I} \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} \mathbf{I} & 0 \\ -\Sigma_{22}^{-1}\Sigma_{21} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \Sigma/\Sigma_{22} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}, \quad (1)$$

where $\Sigma/\Sigma_{22} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$. Next, from equation (1), by taking the determinant of the both sides we have

$$|\Sigma| = \begin{vmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{vmatrix} = |\Sigma/\Sigma_{22}| |\Sigma_{22}|. \quad (2)$$

Second, by inverting both sides of (1) we have,

$$\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{I} & 0 \\ -\Sigma_{22}^{-1}\Sigma_{21} & \mathbf{I} \end{pmatrix} \begin{pmatrix} (\Sigma/\Sigma_{22})^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} - \Sigma_{12}\Sigma_{22}^{-1} \\ 0 & \mathbf{I} \end{pmatrix}.$$

As a result, we have

$$\begin{aligned} p(\mathbf{x}_1, \mathbf{x}_2 | \boldsymbol{\theta}, \Sigma) & \propto \exp \left(-\frac{1}{2} \begin{pmatrix} \mathbf{x}_1 - \boldsymbol{\theta}_1 \\ \mathbf{x}_2 - \boldsymbol{\theta}_2 \end{pmatrix}^T \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{x}_1 - \boldsymbol{\theta}_1 \\ \mathbf{x}_2 - \boldsymbol{\theta}_2 \end{pmatrix} \right) \\ & = \exp \left(-\frac{1}{2} (\mathbf{x}_1 - \boldsymbol{\theta}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\theta}_2))^T (\Sigma/\Sigma_{22})^{-1} (\mathbf{x}_1 - \boldsymbol{\theta}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\theta}_2)) \right) \times \\ & \quad \times \exp \left(-\frac{1}{2} (\mathbf{x}_2 - \boldsymbol{\theta}_2)^T \Sigma_{22}^{-1} (\mathbf{x}_2 - \boldsymbol{\theta}_2) \right). \end{aligned}$$

Moreover, from the identity (2), it can be seen that

$$\begin{aligned} (2\pi)^{(d_1+d_2)/2} |\Sigma|^{1/2} & = (2\pi)^{(d_1+d_2)/2} (|\Sigma/\Sigma_{22}| |\Sigma_{22}|)^{1/2} \\ & = (2\pi)^{d_1/2} |\Sigma/\Sigma_{22}|^{1/2} (2\pi)^{d_2/2} |\Sigma_{22}|^{1/2}. \end{aligned}$$

Hence, we have

$$\begin{aligned} p(\mathbf{x}_1, \mathbf{x}_2 | \boldsymbol{\theta}, \Sigma) & = \frac{1}{(2\pi)^{\frac{d_1}{2}} |\Sigma/\Sigma_{22}|^{\frac{1}{2}}} \\ & \quad \times \exp \left(-\frac{1}{2} (\mathbf{x}_1 - \boldsymbol{\theta}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\theta}_2))^T (\Sigma/\Sigma_{22})^{-1} (\mathbf{x}_1 - \boldsymbol{\theta}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\theta}_2)) \right) \\ & \quad \times \frac{1}{(2\pi)^{\frac{d_2}{2}} |\Sigma_{22}|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (\mathbf{x}_2 - \boldsymbol{\theta}_2)^T \Sigma_{22}^{-1} (\mathbf{x}_2 - \boldsymbol{\theta}_2) \right) \\ & = p(\mathbf{x}_1 | \mathbf{x}_2) p(\mathbf{x}_2). \end{aligned}$$

Therefore, from the above expression, we can see that

$$\begin{aligned}
p(\mathbf{x}_1|\mathbf{x}_2) &= \frac{1}{(2\pi)^{\frac{d_1}{2}} |\boldsymbol{\Sigma}/\Sigma_{22}|^{\frac{1}{2}}} \\
&\times \exp\left(-\frac{1}{2}(\mathbf{x}_1 - \boldsymbol{\theta}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\theta}_2))^T (\boldsymbol{\Sigma}/\Sigma_{22})^{-1}(\mathbf{x}_1 - \boldsymbol{\theta}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\theta}_2))\right) \\
&\sim \text{multivariate-normal}(\boldsymbol{\theta}_{1|2}, \boldsymbol{\Sigma}_{1|2}),
\end{aligned}$$

where

$$\boldsymbol{\theta}_{1|2} = \boldsymbol{\theta}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\theta}_2), \quad \boldsymbol{\Sigma}_{1|2} = \boldsymbol{\Sigma}/\Sigma_{22} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

This completes the proof of the theorem. $\square$

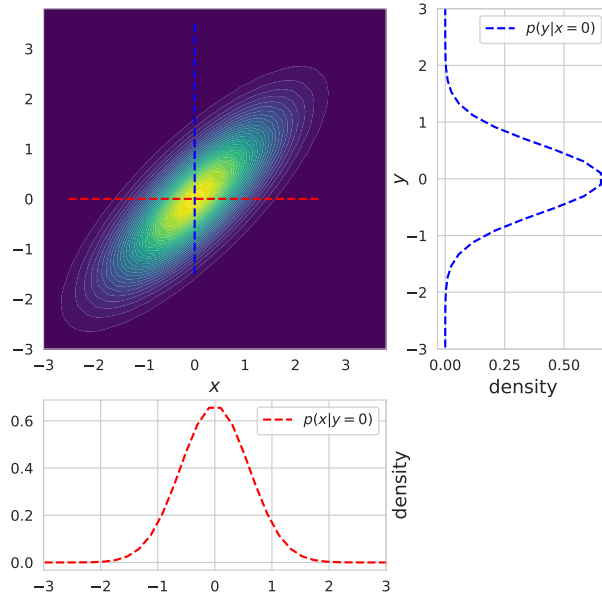


Fig. 2 Conditional distributions

In Figure 2, we illustrate the results of Theorem 3 by plotting the contour of the probability density function of $\mathbf{X} = (X_1, X_2) \sim \text{multivariate-normal}(\boldsymbol{\theta}, \boldsymbol{\Sigma})$ with $\boldsymbol{\theta} = (0, 0)^T$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$, the density functions of $p(x|y=0)$ and $p(y|x=0)$. Clearly, it can be seen that the results are matched to those obtained in Theorem 3.

Exercise 3. Verify that

$$\begin{pmatrix} \mathbf{I} - \Sigma_{12}\Sigma_{22}^{-1} \\ 0 & \mathbf{I} \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} \mathbf{I} & 0 \\ -\Sigma_{22}^{-1}\Sigma_{21} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \Sigma/\Sigma_{22} & 0 \\ 0 & \Sigma_{22} \end{pmatrix},$$

where $\Sigma/\Sigma_{22} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$.

3 Multivariate Normal Bayesian Inference

In this section, it is assumed that the random vector $\mathbf{X} = (X_1, X_2, \dots, X_p)$ has a multivariate normal distribution. That is, the probability density of $\mathbf{X}$ is given by

$$p(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-(\mathbf{x} - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\theta})/2\right), \quad (3)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_p)^T$, $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_p)^T$, and the covariance matrix $\boldsymbol{\Sigma} = (\Sigma_{ij})_{p \times p}$. Given a realized sample $\{\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip}) : i = 1, 2, \dots, n\}$ from the random vector $\mathbf{X} = (X_1, X_2, \dots, X_p)$, our goal is to construct estimates for $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$ by taking the prior distribution knowledge of $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$ and the observed data into account. We will accomplish these goals through several steps which will be outlined in details in the following subsections.

3.1 Bayesian Inference

In probability theory and statistics, Bayes' theorem [2] describes the probability of an event, based on prior knowledge of conditions that might be related to the event. Let us briefly recall the Bayes' theorem here. For any two events A, B with $P(B) \neq 0$,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}. \quad (4)$$

The above formula is the key starting point for Bayesian inference which is a probabilistic method for estimating parameters of wide range of models [12], [10]. Bayesian inference uses Bayes' theorem to update the probability for a hypothesis as more evidence or information becomes available. More specifically,

- $\mathbf{x}$ denotes a data point in general.
- $\boldsymbol{\Gamma}$ denotes the parameter of the data point's distribution. That is, $\mathbf{x} \sim p(\mathbf{x}|\boldsymbol{\Gamma})$.
- $p(\boldsymbol{\Gamma})$ denotes the prior distribution of $\boldsymbol{\Gamma}$.

Bayesian Inference states that

$$p(\boldsymbol{\Gamma}|\mathbf{x}) = \frac{p(\mathbf{x}|\boldsymbol{\Gamma})p(\boldsymbol{\Gamma})}{p(\mathbf{x})}.$$

In the above formula:

- $p(\boldsymbol{\Gamma})$ is our prior belief. It is our knowledge known before the data is observed.
- $p(\mathbf{x}|\boldsymbol{\Gamma})$ is the likelihood. It is a measure of how well the new evidence matches with the prior belief.
- $p(\boldsymbol{\Gamma}|\mathbf{x})$ is the posterior belief. It is our updated knowledge after the data is observed.
- $p(\mathbf{x})$ is the marginal likelihood or the model evidence.

We note that $p(\mathbf{x})$ is often seen as a normalizing constant and ignored. Hence, by ignoring it, we can express the Bayesian inference formula as:

$$p(\boldsymbol{\Gamma}|\mathbf{x}) \propto p(\mathbf{x}|\boldsymbol{\Gamma})p(\boldsymbol{\Gamma}).$$

Bayesian inference estimates the unknown parameter $\boldsymbol{\Gamma}$ using prior information or belief about the variable. Prior information is captured in the form of a distribution $p(\boldsymbol{\Gamma})$. We note that prior distribution $p(\boldsymbol{\Gamma})$ and posterior distribution $p(\boldsymbol{\Gamma}|\mathbf{x})$ can be different in terms of distributions. However, if the posterior distribution and prior distribution both follow the same type of distribution, they are called conjugate distributions. Bayesian inference typically requires computing expectations with respect to the posterior distribution. For instance, the posterior mean, $\int \boldsymbol{\Gamma} p(\boldsymbol{\Gamma}|\mathbf{x}) d\boldsymbol{\Gamma}$, which is often used for point estimation. However, it is often difficult to compute such quantities partly because the posterior distribution $p(\boldsymbol{\Gamma}|\mathbf{x})$ is intractable as the normalizing constant $p(\mathbf{x})$ is usually unknown. For many applications, Bayesian inference is performed using Markov Chain Monte Carlo (MCMC), which estimates expectations with respect to $p(\boldsymbol{\Gamma}|\mathbf{x})$ by sampling from it. Due to technicality of MCMC, we will not go into details in this chapter, the interested reader is invited to refer to [26] for extensive treatments. We instead provide a brief introduction to MCMC in Section 3.4. In the next sections, we will discuss the choice of prior distributions for the mean and covariance matrix of a multivariate normal distribution. We focus mainly on the conjugate priors in this chapter.

3.2 The Posterior Distribution of θ when Σ is known

First note that the joint sampling density or the likelihood function can be written as,

$$\begin{aligned}
p(\mathbf{x}_1, \dots, \mathbf{x}_n | \boldsymbol{\theta}, \boldsymbol{\Sigma}) &= \prod_{i=1}^n \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x}_i - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\theta}) \right) \\
&= \frac{1}{(2\pi)^{np/2} |\boldsymbol{\Sigma}|^{n/2}} \exp \left(-\frac{1}{2} \sum_{i=1}^n [(\mathbf{x}_i - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\theta})] \right) \\
&= \frac{1}{(2\pi)^{np/2} |\boldsymbol{\Sigma}|^{n/2}} \exp \left(-\frac{1}{2} \sum_{i=1}^n \mathbf{x}_i^T \boldsymbol{\Sigma}^{-1} \mathbf{x}_i - \mathbf{x}_i^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\theta} - \boldsymbol{\theta}^T \boldsymbol{\Sigma}^{-1} \mathbf{x}_i + \boldsymbol{\theta}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\theta} \right) \\
&\propto \exp \left(-\frac{1}{2} \boldsymbol{\theta}^T \mathbf{A}_1 \boldsymbol{\theta} + \boldsymbol{\theta}^T \mathbf{b}_1 \right),
\end{aligned}$$

where $\mathbf{A}_1 = n\boldsymbol{\Sigma}^{-1}$, $\mathbf{b}_1 = n\boldsymbol{\Sigma}^{-1}\bar{\mathbf{x}}$, and $\bar{\mathbf{x}} = \left(\frac{1}{n} \sum_{i=1}^n x_{i,1}, \dots, \frac{1}{n} \sum_{i=1}^n x_{i,p} \right)^T$ denotes the sample average. As a result of the above calculation, a convenient conjugate prior distribution for the mean of a multivariate normal distribution is a multivariate normal distribution. Specifically, the prior distribution of $\boldsymbol{\theta}$ is parametrized as follows:

$$p(\boldsymbol{\theta}) = \text{multivariate-normal}(\boldsymbol{\mu}_0, \boldsymbol{\Lambda}_0),$$

for some vector $\boldsymbol{\mu}_0$ and positive definite matrix $\boldsymbol{\Lambda}_0$. We can further simplify the prior distribution as follows:

$$\begin{aligned}
p(\boldsymbol{\theta}) &= \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Lambda}_0|^{1/2}} \exp \left(-\frac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \boldsymbol{\Lambda}_0^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_0) \right) \\
&= \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Lambda}_0|^{1/2}} \exp \left(-\frac{1}{2} \boldsymbol{\theta}^T \boldsymbol{\Lambda}_0^{-1} \boldsymbol{\theta} + \boldsymbol{\theta}^T \boldsymbol{\Lambda}_0^{-1} \boldsymbol{\mu}_0 - \frac{1}{2} \boldsymbol{\mu}_0^T \boldsymbol{\Lambda}_0^{-1} \boldsymbol{\mu}_0 \right) \\
&\propto \exp \left(-\frac{1}{2} \boldsymbol{\theta}^T \boldsymbol{\Lambda}_0^{-1} \boldsymbol{\theta} + \boldsymbol{\theta}^T \boldsymbol{\Lambda}_0^{-1} \boldsymbol{\mu}_0 \right) \\
&= \exp \left(-\frac{1}{2} \boldsymbol{\theta}^T \mathbf{A}_0 \boldsymbol{\theta} + \boldsymbol{\theta}^T \mathbf{b}_0 \right),
\end{aligned}$$

where $\mathbf{A}_0 = \boldsymbol{\Lambda}_0^{-1}$ and $\mathbf{b}_0 = \boldsymbol{\Lambda}_0^{-1} \boldsymbol{\mu}_0$.

Combining the two previous equations for prior and likelihood functions we have the posterior distribution can be found as follows:

$$\begin{aligned}
p(\boldsymbol{\theta} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\Sigma}) &\propto \exp \left(-\frac{1}{2} \boldsymbol{\theta}^T \mathbf{A}_0 \boldsymbol{\theta} + \boldsymbol{\theta}^T \mathbf{b}_0 \right) \times \exp \left(-\frac{1}{2} \boldsymbol{\theta}^T \mathbf{A}_1 \boldsymbol{\theta} + \boldsymbol{\theta}^T \mathbf{b}_1 \right) \\
&= \exp \left(-\frac{1}{2} \boldsymbol{\theta}^T \mathbf{A}_n \boldsymbol{\theta} + \boldsymbol{\theta}^T \mathbf{b}_n \right),
\end{aligned}$$

where

$$\begin{aligned}\mathbf{A}_n &= \mathbf{A}_0 + \mathbf{A}_1 = \mathbf{A}_0^{-1} + n\mathbf{\Sigma}^{-1} \\ \mathbf{b}_n &= \mathbf{b}_0 + \mathbf{b}_1 = \mathbf{A}_0^{-1}\boldsymbol{\mu}_0 + n\mathbf{\Sigma}^{-1}\bar{\mathbf{x}}.\end{aligned}$$

Therefore,

$$p(\boldsymbol{\theta}|\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{\Sigma}) = \text{multivariate-normal}(\boldsymbol{\mu}_n, \mathbf{A}_n), \quad (5)$$

with $\boldsymbol{\mu}_n = (\mathbf{A}_0^{-1} + n\mathbf{\Sigma}^{-1})^{-1}(\mathbf{A}_0^{-1}\boldsymbol{\mu}_0 + n\mathbf{\Sigma}^{-1}\bar{\mathbf{x}})$ and $\mathbf{A}_n = (\mathbf{A}_0^{-1} + n\mathbf{\Sigma}^{-1})^{-1}$. $\boldsymbol{\mu}_n$ and $\mathbf{A}_n$ are called the posterior mean and posterior covariance, respectively. Note that if we look at the posterior covariance $\mathbf{A}_n$ carefully, then it is clear that the posterior precision is the sum of the prior precision and the data precision. Similarly, the posterior mean is a weighted average of the prior mean and the sample mean.

3.3 The Posterior Distribution of Σ when θ is known

Using a matrix algebra identity $\sum_{k=1}^K \mathbf{b}_k^T \mathbf{A} \mathbf{b}_k = \text{tr}(\sum_{k=1}^K \mathbf{b}_k^T \mathbf{A} \mathbf{b}_k) = \text{tr}(\mathbf{B}^T \mathbf{B} \mathbf{A})$ where $\mathbf{B}$ is the matrix whose k th row is $\mathbf{b}_k^T$ and from the calculation in Section 3.2, the likelihood function $p(\mathbf{x}_1, \dots, \mathbf{x}_n | \boldsymbol{\theta}, \mathbf{\Sigma})$ can be written as

$$\begin{aligned}p(\mathbf{x}_1, \dots, \mathbf{x}_n | \boldsymbol{\theta}, \mathbf{\Sigma}) &= \prod_{i=1}^n \frac{1}{(2\pi)^{p/2} |\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\theta})^T \mathbf{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\theta})\right) \\ &= \frac{1}{(2\pi)^{np/2} |\mathbf{\Sigma}|^{n/2}} \exp\left(-\frac{1}{2} \sum_{i=1}^n [(\mathbf{x}_i - \boldsymbol{\theta})^T \mathbf{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\theta})]\right) \\ &= \frac{1}{(2\pi)^{np/2} |\mathbf{\Sigma}|^{n/2}} \exp\left(-\frac{1}{2} \text{tr}\left(\sum_{i=1}^n [(\mathbf{x}_i - \boldsymbol{\theta})^T \mathbf{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\theta})]\right)\right) \\ &\propto |\mathbf{\Sigma}|^{-n/2} \exp\left(-\frac{1}{2} \text{tr}(\mathbf{S}_\theta \mathbf{\Sigma}^{-1})\right),\end{aligned}$$

where

$$\mathbf{S}_\theta = \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T (\mathbf{x}_i - \boldsymbol{\theta}).$$

We note that in the above derivation, we keep Σ as it is assumed to be unknown in this section. Moreover, from the result of the above calculation, we should choose the prior $p(\mathbf{\Sigma})$ that belongs to the same functional family. The corresponding conjugate prior is known as the inverse-Wishart distribution. As said in [23] “The inverse-Wishart distribution ranks next to the (multivariate) normal distribution in order of importance and usefulness in multivariate analysis”. A inverse-Wishart distribution is a multivariate form of an inverse-Gamma distribution. More specifically, for $v_0 > p - 1$ and a positive definite matrix $\mathbf{S}_0 = ((S_0)_{ij})_{p \times p}$, $\mathbf{\Sigma}$ has a inverse-Wishart

distribution with parameters v_0 and $\mathbf{S}_0$ if its probability density function is given as follows:

$$p(\mathbf{\Sigma}) = \left[2^{v_0 p/2} \pi^{(p/2)} |\mathbf{S}_0|^{-v_0/2} \prod_{j=1}^p \Gamma([v_0 + 1 - j]/2) \right]^{-1} \times \\ |\mathbf{\Sigma}|^{-(v_0+p+1)/2} \times \exp[-\text{tr}(\mathbf{S}_0 \mathbf{\Sigma}^{-1})/2].$$

Note that we must have $v_0 > p - 1$ to ensure that $\mathbf{S}_0$ is invertible. We will adopt the notation inverse-Wishart($v_0, \mathbf{S}_0^{-1}$) for an inverse-Wishart distribution with parameters v_0 and $\mathbf{S}_0$. From [20], we have: If $\mathbf{\Sigma} \sim \text{inverse-Wishart}(v_0, \mathbf{S}_0^{-1})$ then

- The mean of $\mathbf{\Sigma}$: $\mathbb{E}(\mathbf{\Sigma}) = \frac{1}{v_0 - p - 1} \mathbf{S}_0$.
- The covariance of elements of $\mathbf{\Sigma} = (\Sigma_{ij})_{p \times p}$ are given by

$$\text{Cov}(\Sigma_{ij}, \Sigma_{kl}) = \frac{2(S_0)_{ij}(S_0)_{kl} + (v_0 - p - 1)((S_0)_{ik}(S_0)_{jl} + (S_0)_{il}(S_0)_{kj})}{(v_0 - p)(v_0 - p - 1)^2(v_0 - p - 3)}.$$

An inverse-Wishart distribution posses many more interesting properties. We will not list them here for the sake of clarity. Interested readers can refer to the excellent books [21, Chapter 4] and [20] for more details.

We assume that the prior distribution of $\mathbf{\Sigma}$ is an Inverse-Wishart distribution. That is, $p(\mathbf{\Sigma}) \sim \text{inverse-Wishart}(v_0, \mathbf{S}_0^{-1})$. Then by combining the the prior distribution and the likelihood function, we obtain the posterior distribution of $\mathbf{\Sigma}$ as follows:

$$p(\mathbf{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\theta}) \propto p(\mathbf{\Sigma}) \times p(\mathbf{x}_1, \dots, \mathbf{x}_n | \boldsymbol{\theta}, \mathbf{\Sigma}) \\ \propto \left(|\mathbf{\Sigma}|^{-(v_0+p+1)/2} \times \exp[-\text{tr}(\mathbf{S}_0 \mathbf{\Sigma}^{-1})/2] \right) \times \left(|\mathbf{\Sigma}|^{-n/2} \exp\left(-\frac{1}{2} \text{tr}(\mathbf{S}_\theta \mathbf{\Sigma}^{-1})\right) \right) \\ = |\mathbf{\Sigma}|^{-(v_0+n+p+1)/2} \exp(-\text{tr}([\mathbf{S}_0 + \mathbf{S}_\theta] \mathbf{\Sigma}^{-1})/2).$$

Therefore, the posterior distribution for variance when the mean is known for a multivariate normal model is an inverse-Wishart distribution. That is,

$$p(\mathbf{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\theta}) \sim \text{inverse-Wishart}(v_0 + n, [\mathbf{S}_0 + \mathbf{S}_\theta]^{-1}). \quad (6)$$

It can be seen that

$$\mathbb{E}(\mathbf{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\theta}) = \frac{v_0 - p - 1}{v_0 + n - p - 1} \frac{1}{v_0 - p - 1} \mathbf{S}_0 + \frac{n}{v_0 + n - p - 1} \frac{1}{n} \mathbf{S}_\theta,$$

which can be interpreted as a weight sum of the prior mean $\frac{1}{v_0 - p - 1} \mathbf{S}_0$ and the unbiased estimator of the true covariance matrix.

3.4 Gibbs Sampling

To do Bayesian inference, one needs to find the joint conditional distribution $p(\boldsymbol{\theta}, \boldsymbol{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n)$. However, from the calculations in Section 3.2 and Section 3.3, it is difficult to find the joint conditional distribution $p(\boldsymbol{\theta}, \boldsymbol{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n)$. We however are able to find the conditional distributions $p(\boldsymbol{\theta} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\Sigma})$ and $p(\boldsymbol{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\theta})$. Hence we will use these two conditional distributions to do the Bayesian inference through the Gibbs sampling algorithm. Roughly speaking, the Gibbs sampler proceeds as follows: Let $\boldsymbol{\theta}^{(0)}$ and $\boldsymbol{\Sigma}^{(0)}$ be two initial starting values (guesses) of $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$, then we sample $\boldsymbol{\theta}^{(1)} | \boldsymbol{\Sigma}^{(0)}$, then sample $\boldsymbol{\Sigma}^{(1)} | \boldsymbol{\theta}^{(1)}$, and so on. More precisely,

1. Set $(\boldsymbol{\theta}^{(0)}, \boldsymbol{\Sigma}^{(0)})$ be the initial values.
2. Sample $\boldsymbol{\theta}^{(1)} \sim p(\boldsymbol{\theta} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\Sigma}^{(0)})$
3. Sample $\boldsymbol{\Sigma}^{(1)} \sim p(\boldsymbol{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\theta}^{(1)})$
4. Sample $\boldsymbol{\theta}^{(2)} \sim p(\boldsymbol{\theta} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\Sigma}^{(1)})$
5. Sample $\boldsymbol{\Sigma}^{(2)} \sim p(\boldsymbol{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\theta}^{(2)})$
6. And so on.

Each iteration in the Gibbs sampling algorithm is also referred to as a *sweep* or *scan*. Note that when updating one variable, we always use the most recent value of the other variable (even in the middle of an iteration). This procedure defines a sequence of pairs of random variables:

$$(\boldsymbol{\theta}^{(0)}, \boldsymbol{\Sigma}^{(0)}), (\boldsymbol{\theta}^{(1)}, \boldsymbol{\Sigma}^{(1)}), (\boldsymbol{\theta}^{(2)}, \boldsymbol{\Sigma}^{(2)}), \dots, (\boldsymbol{\theta}^{(i)}, \boldsymbol{\Sigma}^{(i)}), \dots$$

which has the property of being a *Markov chain*, that is, the conditional distribution of $(\boldsymbol{\theta}^{(i)}, \boldsymbol{\Sigma}^{(i)})$ given all of the previous pairs depends only on $(\boldsymbol{\theta}^{(i-1)}, \boldsymbol{\Sigma}^{(i-1)})$. Under suitable conditions [25], it can be showed that for a function $g(\boldsymbol{\theta}, \boldsymbol{\Sigma})$ with $\mathbb{E}[g(\boldsymbol{\theta}, \boldsymbol{\Sigma})] < \infty$ then

$$\frac{1}{M} \sum_{i=1}^M g(\boldsymbol{\theta}^{(i)}, \boldsymbol{\Sigma}^{(i)}) \rightarrow \mathbb{E}[g(\boldsymbol{\theta}, \boldsymbol{\Sigma})] \quad \text{as } M \rightarrow \infty.$$

As a result, for a sufficient large M ,

$$\mathbb{E}[g(\boldsymbol{\theta}, \boldsymbol{\Sigma})] \approx \frac{1}{M} \sum_{i=1}^M g(\boldsymbol{\theta}^{(i)}, \boldsymbol{\Sigma}^{(i)}).$$

This approach is often referred to as *Markov chain Monte Carlo*. To start, pick an arbitrary starting value for each parameter, we will use the sample means $\boldsymbol{\theta}^{(0)} = \hat{\boldsymbol{\mu}}_{mle}$ and sample covariance matrix $\boldsymbol{\Sigma}^{(0)} = \hat{\boldsymbol{\Sigma}}_{mle}$ (recall Theorem 2) as the starting values. Recall that $\mathbf{v}_n = \mathbf{v}_0 + n$ and $\mathbf{S}_n = \mathbf{S}_0 + \mathbf{S}_\theta$. We note that the initial values of the (hyper) parameters $\boldsymbol{\mu}_0, \boldsymbol{\Lambda}_0, \mathbf{v}_0, \mathbf{S}_0$ very much depend on data. We list the possible initial values of the (hyper) parameters as follows:

1. $\boldsymbol{\mu}_0 = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i, \boldsymbol{\Lambda}_0 = \hat{\boldsymbol{\Sigma}}_{mle}$.

2. $v_0 = p + 2$, $\mathbf{S}_0 = \text{diag}(\hat{\boldsymbol{\Sigma}}_{mle})$.

One can refer to [21] and [12] for more discussions on the initial values of the (hyper) parameters. We summarize the Gibbs sampling algorithm in Algorithm 1.

Algorithm 1: Gibbs sampling algorithm

Require: $\boldsymbol{\mu}_0, \mathbf{A}_0, v_0, \mathbf{S}_0, \boldsymbol{\theta}^{(0)}, \boldsymbol{\Sigma}^{(0)}$

Require: Number of iterations M

- 1: **for** $i = 1, 2, \dots, M$ **do**
 - 2: sample $\boldsymbol{\theta}^{(i)}$:
 1. compute $\boldsymbol{\mu}_n$ and $\mathbf{A}_n$ from $\mathbf{x}_1, \dots, \mathbf{x}_n$ and $\boldsymbol{\Sigma}^{(i-1)}$
 2. sample $\boldsymbol{\theta}^{(i)} \sim p(\boldsymbol{\theta} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\Sigma}^{(i-1)})$ using (5)
 - 3: sample $\boldsymbol{\Sigma}^{(i)}$:
 1. compute $\mathbf{S}_n$ from $\mathbf{x}_1, \dots, \mathbf{x}_n$ and $\boldsymbol{\theta}^{(i)}$: $\mathbf{S}_n = \mathbf{S}_0 + \mathbf{S}_{\boldsymbol{\theta}^{(i)}}$
 2. sample $\boldsymbol{\Sigma}^{(i)} \sim p(\boldsymbol{\Sigma} | \mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\theta}^{(i)})$ using (6)
 - 4: **end for**
-

We note that the program language *R* provides two built-in functions *rmvnorm* and *rwish* which can be used to sample multivariate normal random variables and Wishart random variables, respectively.

4 Bayesian Data Imputation

4.1 Types of Missing Values

Collecting data is often a messy process. Incomplete data sets are very common when working with real world data. The data could be missing due to wide variety of reasons such as a page on a survey being easy to skip, combining data sets with different variables, or data loss due to storage corruption. According to [27], there are four types of missing data: missing completely at random, missing at random, missing not at random, and structurally missing.

Missing completely at random means that the missing data points don't follow a pattern and are completely independent from the variables. For example, this could occur naturally if each subject is given a random set of survey questions to complete. This would cause data to be missing completely at random independent of how the other questions were answered. However, it is usually not the case that data are missing completely at random, as there is usually an underlying pattern that would make certain variables more or less likely to be missing.

The next type of missing data is missing at random. This is when there is a pat-

tern to the missing data, but the missing values can be predicted from the existing data. This means that the missing data occurs with a wide range of existing values in the same row. For example, a sensor may have malfunctioned for several minutes causing there to be a gap in the data. By using the readings of the other sensor and previous data the missed readings can be accurately predicted.

Missing not at random is similar to missing at random except that the missing data cannot be accurately predicted. This means that there is a lack of data from key subgroups in the data. For example, a survey might ask the income of the subject. Those with low income may be less likely to answer causing the average income to appear far higher than it actually is.

The final type of missing data is structurally missing data. This is data that is missing for a reason done on purpose by the researcher. For example, the income from employment for people without jobs would be null. Data engineering can also cause missing data. For example, when working with time series data there may be dates that don't have any data causing the appearance of "holes" in the data.

4.2 Data Imputation

In this section, assume that we can collect an iid data set $\{\mathbf{x}_i : i = 1, 2, \dots, n\}$ such that $\mathbf{x}_i \sim \text{multivariate-normal}(\boldsymbol{\theta}, \boldsymbol{\Sigma})$. However, it is also assumed that $x_{ij} = NA$ for some $i \in \{1, 2, \dots, n\}, j \in \{1, 2, \dots, p\}$. This is often the case for most of data set including medical data, economic data, etc. The major goal of this section is to "fill in" those missing values using a Bayesian approach. There are several methods to deal with the missing values when they are missing completely at random or missing at random. Listwise deletion is a rudimentary imputation method that involves deleting every row with a missing value. It is a poor imputation method because the deleted rows contain lots of information that is now lost. Because of this, listwise deletion is almost never the best solution for dealing with missing data. Mean/median imputation is slightly better than listwise deletion, but it still has significant drawbacks. It involves imputing the missing values with the mean or median of the column. This retains the missing information by not deleting rows like listwise deletion, but it doesn't take into account the relationship between the variables. This causes the confidence interval for predictions to be much wider than they would be with a better imputation method. Mean/median imputation is very simple way to analyze data, but there are more sophisticated options better suited for more in depth analysis.

A better approach is to use a Bayesian method to impute the data. The Bayesian approach uses Gibbs sampling to take the mean of the column and adjust it based on the other values in the row and the relationship between the variables. To start, let $\mathbf{O}_i = (O_{i,1}, \dots, O_{i,p})^T$ be a binary list populated with zeros and ones such that

$O_{i,j} = 1$ means that the value $X_{i,j}$ is present and $O_{i,j} = 0$ means that the value $X_{i,j}$ is missing. In the following, let $\mathbf{o}_i, \mathbf{x}_{ij}$ be realizations of $\mathbf{O}_i, X_{ij}$, respectively. Using the conditional probability identity (4), the sampling probability for the data for subject i is then as follows:

$$\begin{aligned} p(\mathbf{o}_i, \mathbf{x}_{i,j} : o_{i,j} = 1 | \boldsymbol{\theta}, \boldsymbol{\Sigma}) &= p(\mathbf{o}_i) \times p(\mathbf{x}_{i,j} : o_{i,j} = 1 | \boldsymbol{\theta}, \boldsymbol{\Sigma}) \\ &= p(\mathbf{o}_i) \times \int \{p(x_{i,1}, \dots, x_{i,p} | \boldsymbol{\theta}, \boldsymbol{\Sigma}) \prod_{x_{i,j}: o_{i,j}=0} dx_{i,j}\}. \end{aligned}$$

This means that the sampling probability for data for subject i is $p(\mathbf{o}_i)$ times the marginal probability of the existing variables after integrating out the unobserved values. When using a multivariate normal model for Bayesian inference, $\int p(x_{i,1}, \dots, x_{i,p} | \boldsymbol{\theta}, \boldsymbol{\Sigma}) \prod_{x_{i,j}: o_{i,j}=0} dx_{i,j}$ has no closed form solution. Therefore, we will use Gibbs sampling to find estimates for the missing values. Let $\mathbf{X}$ be an $n \times p$ matrix that holds all of the data obtained and $\mathbf{O}$ be an $n \times p$ matrix in which $o_{i,j} = 1$ if $X_{i,j}$ is observed and $o_{i,j} = 0$ if $X_{i,j}$ is not observed. We will then split the matrix $\mathbf{X}$ into two disjoint parts: $\mathbf{X} = \mathbf{X}_{obs} \cup \mathbf{X}_{miss}$ with

$$\begin{aligned} \mathbf{X}_{obs} &= \{x_{i,j} : o_{i,j} = 1\} \\ \mathbf{X}_{miss} &= \{x_{i,j} : o_{i,j} = 0\}. \end{aligned}$$

From our observed data we want to obtain $p(\boldsymbol{\theta}, \boldsymbol{\Sigma}, \mathbf{X}_{miss} | \mathbf{X}_{obs})$ which is the posterior distribution of the data set and the missing values. A Gibbs sampler can be used to approximate this distribution and the missing values by adding a step to the previous section's Gibbs sampler, Section 3.4: Giving starting values (guesses) for $\boldsymbol{\theta}^{(0)}, \mathbf{X}_{miss}^{(0)}, \boldsymbol{\Sigma}^{(0)}$, we generate the next values using the following iterative scheme,

1. sample $\boldsymbol{\theta}^{(s+1)}$ from $p(\boldsymbol{\theta} | \mathbf{X}_{obs}, \mathbf{X}_{miss}^{(s)}, \boldsymbol{\Sigma}^{(s)})$
2. sample $\boldsymbol{\Sigma}^{(s+1)}$ from $p(\boldsymbol{\Sigma} | \mathbf{X}_{obs}, \mathbf{X}_{miss}^{(s)}, \boldsymbol{\theta}^{(s+1)})$
3. sample $\mathbf{X}_{miss}^{(s+1)}$ from $p(\mathbf{X}_{miss} | \mathbf{X}_{obs}, \boldsymbol{\theta}^{(s+1)}, \boldsymbol{\Sigma}^{(s+1)})$.

For steps one and two, $\mathbf{X}_{obs}$ and $\mathbf{X}_{miss}$ combine to form $\mathbf{X}$. This means that $p(\boldsymbol{\theta} | \mathbf{X}_{obs}, \mathbf{X}_{miss}^{(s)}, \boldsymbol{\Sigma}^{(s)})$ is multivariate normally distributed as in (5). Similarly, from (6) $p(\boldsymbol{\Sigma} | \mathbf{X}_{obs}, \mathbf{X}_{miss}^{(s)}, \boldsymbol{\theta}^{(s+1)})$ has an inverse-Wishart distribution. As a result, these steps can be completed in the same way as done in the previous section where $\boldsymbol{\theta}$ is sampled from a multivariate normal distribution and $\boldsymbol{\Sigma}$ is sampled from an inverse-Wishart distribution. Step 3 is sampled as follows:

$$\begin{aligned} p(\mathbf{X}_{miss} | \mathbf{X}_{obs}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) &\propto p(\mathbf{X}_{miss}, \mathbf{X}_{obs} | \boldsymbol{\theta}, \boldsymbol{\Sigma}) \\ &= \prod_{i=1}^n p(\mathbf{x}_{i,miss}, \mathbf{x}_{i,obs} | \boldsymbol{\theta}, \boldsymbol{\Sigma}) \\ &\propto \prod_{i=1}^n p(\mathbf{x}_{i,miss} | \mathbf{x}_{i,obs}, \boldsymbol{\theta}, \boldsymbol{\Sigma}). \end{aligned} \tag{7}$$

For each i we need to sample the missing values conditionally to the observed elements in the row. Next, from Theorem 3, we have

$$p(\mathbf{x}_{i,miss} | \mathbf{x}_{i,obs}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) \sim \text{multivariate-normal}(\boldsymbol{\theta}_{miss|obs}, \boldsymbol{\Sigma}_{miss|obs}),$$

where

$$\begin{aligned} \boldsymbol{\theta}_{miss|obs} &= \boldsymbol{\theta}_{[miss]} + \boldsymbol{\Sigma}_{[miss,obs]} (\boldsymbol{\Sigma}_{[obs,obs]})^{-1} (\mathbf{x}_{[obs]} - \boldsymbol{\theta}_{[obs]}) \\ \boldsymbol{\Sigma}_{miss|obs} &= \boldsymbol{\Sigma}_{[miss,miss]} - \boldsymbol{\Sigma}_{[miss,obs]} (\boldsymbol{\Sigma}_{[obs,obs]})^{-1} \boldsymbol{\Sigma}_{[obs,miss]}. \end{aligned}$$

Note that here we have used the notation $\boldsymbol{\Sigma}_{[\mathbf{A},\mathbf{B}]}$ to denote the matrix $\boldsymbol{\Sigma}$ with rows in $\mathbf{A}$ and columns in $\mathbf{B}$. In the formulae above, we can see that the missing values are calculated by taking the unconditional mean and then modifying it by looking at the observed values. Similarly, the missing data covariance matrix is equal to the unconditional covariance matrix with a little bit subtracted from it. This makes sense, as having more information about the variables should decrease our uncertainty of their actual values. We summarize the implementation process in Algorithm 2.

Algorithm 2: Bayesian Imputation Algorithm

Require: $\boldsymbol{\mu}_0, \boldsymbol{\Lambda}_0, \nu_0, \mathbf{S}_0, \boldsymbol{\theta}^{(0)}, \boldsymbol{\Sigma}^{(0)}, \mathbf{X}_{miss}^{(0)}$

Require: Number of iterations M

- 1: **for** $s = 1, 2, \dots, M$ **do**
 - 2: Sample $\boldsymbol{\theta}^{(s+1)}$ from $p(\boldsymbol{\theta} | \mathbf{X}_{obs}, \mathbf{X}_{miss}^{(s)}, \boldsymbol{\Sigma}^{(s)})$ using (5)
 - 3: Sample $\boldsymbol{\Sigma}^{(s+1)}$ from $p(\boldsymbol{\Sigma} | \mathbf{X}_{obs}, \mathbf{X}_{miss}^{(s)}, \boldsymbol{\theta}^{(s+1)})$ using (6)
 - 4: Sample $\mathbf{X}_{miss}^{(s+1)}$ from $p(\mathbf{X}_{miss} | \mathbf{X}_{obs}, \boldsymbol{\theta}^{(s+1)}, \boldsymbol{\Sigma}^{(s+1)})$ using (7)
 - 5: **end for**
-

In the next sections, we will provide several examples to demonstrate the power of the Bayesian data imputation.

4.3 Example: Iris Flowers

In this section, we provide some examples to demonstrate the idea of Bayesian imputation. We note that one can apply the idea of Bayesian imputation to any financial data set. The data set that will be imputed is the Iris flower data set [8]. The iris data set is a 150 iris flower measurements done by Ronald Fisher in his 1936 paper “The use of multiple measurements in taxonomic problems”. There are three flowers: Iris setosa, Iris virginica, and Iris versicolor. Each flower has four measurements: sepal length, sepal width, petal length, and petal width. From these four measurements it is possible to create a highly accurate model for predicting the type of iris flower. In Figure 3, we plot the correlations among the variables in the data set. We note that for later comparison, the Iris data set is a complete data set with no missing values. This allows us to delete values at random and then compare the imputed

values to the original to gauge the effectiveness of the imputation. The code for this section can be found at https://github.com/nducduy/Bayesian_Imputation/blob/main/Imputation_main_imputation.Rmd.

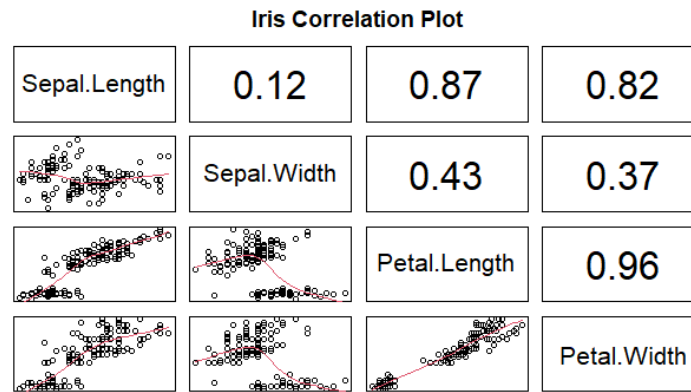


Fig. 3 Correlations among Iris's variables

As can be seen in Figure 3, the Sepal Length, Petal Length, and Petal Width values are all highly correlated. This should make it easy to impute these values. The Sepal Width value is less correlated with the others, so it will likely be more difficult to impute.

Now to test the efficacy of Bayesian imputation, some values must be deleted from the data set. This was done in a way to create data that is missing completely at random. 20% of the values in each column were deleted randomly giving the following pattern of missing data. The distribution of missing values are reported in Figure 4. In Figure 4, the purple represents values that are missing and blue represents values that are not missing. The number on top is the number of rows that follow that pattern for missing data.

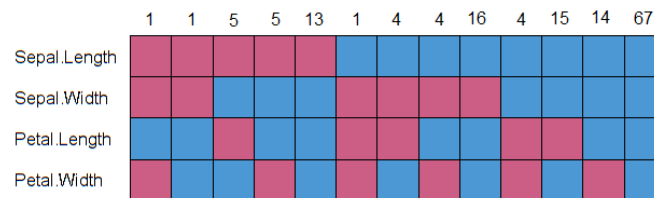


Fig. 4 Distribution of missing values in Iris data set

Gibbs sampling was then used to implement Bayesian imputation on the missing data. We use the posterior mean of the $\mathbf{X}_{miss}$ to fill in the missing values in the data set. Since the true data is known so we can compare the imputed data with the true data. In Figure 5, we report the true values of missing values versus their Bayesian posterior mean for all variables under considerations. As expected, the Bayesian approach performs very well for this data set.

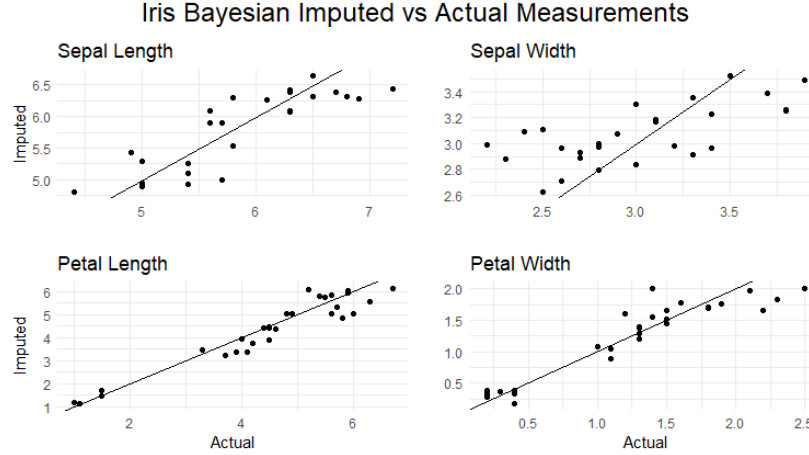


Fig. 5 True values of missing values versus their posterior means

To further our comparison, we use three methods for dealing with missing data mentioned above (listwise deletion, mean imputation, and Bayesian imputation) were then compared against the original data set. The metrics for comparison are mean absolute deviation of the imputed values from the original values and the success rate of the classification algorithms such as random forest models (see [13]) trained on the data set. We use the mean absolute deviation (MAD) as follows: Let n^{miss} denote the total missing values in the i th column and $x_i^{imputed}$ is the imputed value of the true x_i , then

$$MAD = \frac{1}{n^{miss}} \sum_{i=1}^{n^{miss}} |x_i - x_i^{imputed}|.$$

In Table 1, we report the mean absolute deviation of two approaches: Mean imputation and Bayesian imputation. It can be seen that the Bayesian approach has much smaller mean absolute deviation, which means that it performs better than the mean imputation approach.

Next, we use original data set and the imputed data set obtained from listwise deletion, mean imputation and Bayesian imputation algorithms to perform classification using all variables. We choose a logistic regression model and a random forest model to classify the flower species. The model accuracy for the eight models created is in Table 2.

Table 1 Mean absolute deviation: Mean imputation versus Bayesian imputation

Imputation Method	Sepal Length	Sepal Width	Petal Length	Petal Width
Mean	0.59	0.40	1.56	0.55
Bayesian	0.32	0.30	0.36	0.17

Table 2 Classification Accuracy

Model	Original Data	Listwise Deletion	Mean Imputation	Bayesian Imputation
Logistic Regression	0.96	0.89	0.87	0.94
Random Forest	0.95	0.83	0.91	0.95

The results from Table 2 show that the Bayesian imputation method is by far the best approach to dealing with missing data. The Bayesian data set has almost the same predictive power as the original data set. The stronger the correlation between the variables, the better Bayesian imputation performs compared to the other methods. This can be seen in the variable with the lowest correlation, sepal width, for which Bayesian imputation is only marginally better than mean imputation. The data set that was imputed using a Bayesian method has almost the same predictive power of the original data set. The other methods fall far behind in predictive accuracy.

4.4 Value at Risk example

In this section, we will provide an example to show the power of Bayesian data imputation in risk management. Suppose that X is a real-valued random variable representing the risky asset's losses over a fixed horizon, and suppose that X has the density function f and distribution function F . For a confidence level $\alpha \in (0, 1)$, define the *Value-at-Risk* (VaR) as

$$VaR_{\alpha}(X) := \inf\{x \in \mathbb{R} : \mathbb{P}(X \leq x) \geq \alpha\} = \inf\{x \in \mathbb{R} : F(x) \geq \alpha\} = F^{-1}(\alpha), \quad (8)$$

which is simply the α -quantile of F . VaR plays important roles in risk management. It has two main objectives: i) VaR provides a measure of risk; ii) VaR can be used to determine an adequate level of capital reserves that matches the current level of risk.

Exercise 4. Note that in this case $X \sim N(\mu, \sigma^2)$, then for $\alpha \in (0, 1)$,

$$\begin{aligned}
VaR_\alpha(X) &= \inf\{x \in \mathbb{R} : \mathbb{P}(X \leq x) \geq \alpha\} \\
&\Leftrightarrow \alpha = \mathbb{P}(X \leq VaR_\alpha(X)) \\
&\Leftrightarrow \alpha = \mathbb{P}(\mu + \sigma N(0, 1) \leq VaR_\alpha(X)) \\
&\Leftrightarrow \alpha = \mathbb{P}(N(0, 1) \leq (VaR_\alpha(X) - \mu)/\sigma).
\end{aligned} \tag{9}$$

Let $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$ and $\Phi^{-1}(x)$ denotes its inverse. Use the calculation in (9) to show that

$$VaR_\alpha(X) = \mu + \sigma \Phi^{-1}(\alpha). \tag{10}$$

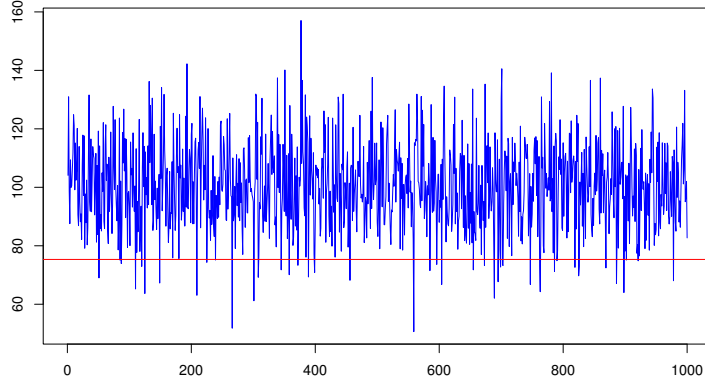


Fig. 6 Normal Value at Risk with $\alpha = 0.05$.

In Figure 6, we simulate and plot (in blue) 1000 random values from $X \sim N(100, 15^2)$. The $VaR_{0.05}(X) = 75.3272$ is plotted in red.

Assume that $\mathbf{X} = (X_1, X_2, \dots, X_p)^T \sim \text{multivariate-normal}(\boldsymbol{\theta}, \boldsymbol{\Sigma})$. Let $S = \mathbf{w}^T \mathbf{X}$, where $\mathbf{w} = (w_1, w_2, \dots, w_p)^T$ with $\sum_{i=1}^p w_i = 1$. It can be showed that $S \sim N(\boldsymbol{\theta}^T \mathbf{w}, \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w})$.

Exercise 5. Prove that

$$S \sim N(\boldsymbol{\theta}^T \mathbf{w}, \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}).$$

As a result, from (10) that

$$VaR_\alpha(S) = \boldsymbol{\theta}^T \mathbf{w} + \Phi^{-1}(\alpha) \sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}. \tag{11}$$

One can cast this problem as a investment problem where an agent can invest a proportion of her wealth (w_i) into the i th risky asset (X_i) and wants to estimate the

value at risk of the total portfolio. From equation (11) it can be seen that one need to estimate $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$ from a historical data in order to estimate $VaR_\alpha(S)$. Some values may be missing. Hence, it is a perfect place to use Bayesian data imputation to estimate $VaR_\alpha(S)$. In this example, we choose $\mathbf{w} = (1/2, 1/2)$ and we will simulate 1000 values of $\mathbf{X} \sim p(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\Sigma})$ with $\boldsymbol{\theta} = (0, 0)^T$, $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$. We then will random remove 20% values of $\mathbf{X}$. Next we use the mean imputation and Bayesian imputation to estimate $VaR_{0.05}(S)$. The results are reported in Table 3.

Table 3 Value at Risk of a portfolio

True VaR	Mean imputed VaR	Bayesian imputed VaR
-1.5604	-1.3874	-1.6298

As we can see from Table 3 that even though we have randomly removed 20% of the data set, the value at risk obtained from using Bayesian imputation is closely matched to the true value at risk.

In the remainder of this section, we aim to present two research projects that will assist readers in gaining a deeper understanding of Bayesian data imputation techniques.

Research Project 1. The concept of Value at Risk for multivariate distributions have been considered in literature. The interested reader can refer to [7] and [22] for details. In this example, we mainly focus on a multivariate normal distribution case. Assume that $\mathbf{X} = (X_1, X_2, \dots, X_p) \sim \text{multivariate-normal}(\boldsymbol{\theta}, \boldsymbol{\Sigma})$ present the dynamics of p different assets. In this case, $\boldsymbol{\theta}$ captures the average level of returns while the covariance matrix $\boldsymbol{\Sigma}$ captures the interactions among the assets. Assume that the current value of the portfolio is V and $\mathbf{w} = (w_1, \dots, w_p)$ is the weight vector corresponding to p assets. The $VaR_\alpha(\mathbf{X})$ can be estimated as (see [32]),

$$VaR_\alpha(\mathbf{X}) = V(\mathbf{w}^T \boldsymbol{\theta} - \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \Phi^{-1}(1 - \alpha)). \quad (12)$$

1. Verify the formula (12).
2. Simulate 5000 values of $\mathbf{X}$ using the parameters of your choice. Next randomly remove 25% of the data.
3. Write a program using Algorithm 2 to impute the missing values and then estimate $VaR_{0.05}(\mathbf{X})$ from the imputed data set.
4. Use the *mice* package of [30] to impute the missing value and estimate $VaR_{0.05}(\mathbf{X})$ from the imputed data set.

Research Project 2. Consider the “Hitters” data set from the *ISLR* R package, [13]. One can download the data set from the link: <https://www.kaggle.com/datasets/mathchi/hitters-baseball-data>. It is a data frame which contains major league baseball data from the 1986 and 1987 seasons. It has 322 rows and 20 columns including player’s names, number of times at bat (AtBat), number of hits (Hits), etc. Note that in this project, we would like to estimate the player’s salary using, for example, AtBat, Hits, and other variables. Also, we would like to consider the effect of data data imputation on estimating players’ salaries.

- Please suggest a model of your preference for estimating a player’s salary. To develop an effective model, it would be beneficial to conduct some preliminary analysis.
- Randomly remove, say 20%, of the data.
- Write a program using Algorithm 2 to impute the missing values and then use the imputed data set to estimate the salary of a play using the model considered previously. You can use, for example, a linear regression model or a random forest model.
- Find the Root Mean Squared Error (RMSE) for each model using the original data set, the listwise deleted data set, and the data set with Bayesian imputation.
- Use the *mice* package of [30] to impute the missing value and estimate the RMSE for each model and each data set considered in the previous question.

5 Conclusion

Data sets are often incomplete due to a variety of reasons. In order to utilize all of the existing data, imputation methods are needed. As seen above, Bayesian imputation offers far better results than the traditional methods of listwise deletion and mean/median imputation. Bayesian imputation is an essential tool to working with incomplete data sets. There are a few downsides to Bayesian such as a slow run time and the need for variable correlation, but for most cases the benefits far outweigh the costs. The previous examples offer strong evidence that Bayesian imputation is very powerful and useful for increasing the predictive power of an incomplete data set.

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